B.Tech Project Report

on

Mouse Dynamics-Based Intrusion Detection

Using Recurrence Plots and Vision Transformers

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Abstract

This report presents the implementation and evaluation of a **mouse dynamics-based continuous authentication system** using **Asymmetric Recurrence Plots (ARP)** and a lightweight **Vision Transformer (ViT)**. The system is designed to detect impostors by analysing the unique spatial patterns of each user's mouse movement, encoded as recurrence plot images and classified by a transformer-based neural network with Gradient Harmonizing Mechanism (GHM) loss. The model was trained and evaluated on the **DFL dataset** (21 users), achieving a mean AUC of **0.7800** and a mean EER of **0.2756** in a single-epoch training regime on CPU. The implementation faithfully reproduces the architecture described in the reference paper (Equations 1–21), including efficient multi-head self-attention, GHM loss, and ARP generation. A performance gap relative to the paper (AUC=0.9919, EER=0.0220) is attributed to reduced training epochs and hardware constraints.

Keywords: Mouse Dynamics, Intrusion Detection, Recurrence Plots, Vision Transformer, Continuous Authentication, GHM Loss, Cybersecurity

Chapter 1: Introduction

Continuous authentication is an active area of research in behavioural biometrics, aiming to verify user identity beyond the initial login by monitoring ongoing interaction patterns. Mouse dynamics — the spatial and temporal characteristics of cursor movement — provide a rich, non-intrusive signal for this purpose. Unlike static biometrics (fingerprints, iris scans), mouse behaviour can be captured transparently in the background without disrupting user workflow.

Traditional anomaly detection approaches for mouse dynamics rely on handcrafted statistical features such as velocity, acceleration, curvature, and click duration. While effective, these require domain expertise for feature engineering and may not generalise well across diverse user populations and interaction contexts.

This project implements a **deep learning pipeline** that transforms raw mouse coordinate sequences into 2D visual representations (Recurrence Plots) and feeds them to a Vision Transformer classifier. Key innovations include:

- **Asymmetric Recurrence Plots (ARP):** A compact 300×300 image encoding spatial recurrence from mouse trajectories (Equation 4).
- **Efficient Multi-Head Self-Attention:** $O(N \cdot dK)$ complexity instead of standard $O(N^2)$, enabling faster inference (Equations 15–16).
- **Gradient Harmonizing Mechanism (GHM) Loss:** Down-weights easy samples and focuses training on hard-to-classify examples (Equations 17–21).
- **On-the-fly Plot Generation:** Generates recurrence plots during training instead of storing pre-computed images, reducing memory from ~202 GB to ~2.7 GB.

Chapter 2: Methodology

2.1 Dataset and Configuration

The DFL (DFL Mouse Dataset) was used for all experiments. It contains mouse event logs from 21 users across multiple sessions. The dataset was structured as follows:

Data Location: `/content/drive/MyDrive/mouse_project/dfl`

Structure: user_ID / session_N.csv, with columns for timestamp, x, y coordinates.

Training Configuration:

Parameter	Value	Description
M (segment length)	600	Mouse events per segment
Epsilon (ε)	0.15	Recurrence threshold (DFL best)
Plot Type	ARP	Asymmetric Recurrence Plot
Image Size	300×300	ARP output dimensions
Patch Size	15×15	ViT patch dimension P
Embed Dim	225 (=P²)	Linear patch embedding size
Num Heads	3	Attention heads h
Num Layers	3	Transformer encoder blocks L
MLP Size	512	Feed-forward hidden size
Head Dim	75	dQ = dK = dV per head
Dropout	0.1	Regularisation rate
Epochs	1	Training epochs (limited by CPU)
Batch Size	64	Samples per gradient update
Learning Rate	0.001	Initial LR; decays at 60,80
GHM Bins	10	Gradient histogram bins
Impostor Cap (train)	5,000	Max impostor segments per user
Impostor Cap (test)	2,000	Max impostor segments for eval

2.2 Dataset Statistics

After loading and standardising all 21 users, the dataset yielded the following segment counts:

User	Train Segments	Test Segments	Note
User1	666	82	

User	Train Segments	Test Segments	Note
User2	6,098	761	1 session had CSV error
User3	187	23	
User4	183,791	22,973	Largest user
User5	64,785	8,099	
User6	17,303	2,163	
User7	17,590	2,198	
User8	3,029	378	
User9	2,060	257	
User10	2,105	263	
User11	830	103	
User12	26,193	3,275	
User13	3,899	488	
User14	11,403	1,424	
User15	48,625	6,075	
User16	8,902	1,112	
User17	395	48	Smallest user
User18	846	107	
User19	71,570	8,946	
User20	12,288	1,534	
User21	16,833	2,105	

Total: 499,398 train segments, 62,414 test segments across 21 users.

2.3 Recurrence Plot Generation (Equations 1–4)

Mouse coordinate sequences are transformed into 2D grayscale images via recurrence analysis. The pipeline implements four equations from the reference paper:

1. **Eq.1 — Distance Matrix:** $d(i,j) = ||p_i - p_j||_2$ for all coordinate pairs in a segment of length M.
2. **Eq.2 — Recurrent Points:** $avg(p_i) = 1/(M-1) \times \Sigma_{j \neq i} d(i,j)$; point i is recurrent if $avg(p_i) < \varepsilon$.
3. **Eq.3 — Recurrence Matrix:** $r(i,j) = d(i,j)$ if both p_i and p_j are recurrent; else $r(i,j) = \varepsilon$.
4. **Eq.4 — Asymmetric Recurrence Plot:** Split M coordinates into two halves. Build SRP_1 (first half) and SRP_2 (second half). ARP = upper-triangle(SRP_1) $\oplus$ lower-triangle(SRP_2). Output: (M/2) × (M/2) = 300×300 grayscale image.

The key advantage of ARP over SRP is a 4× reduction in image area (300×300 vs 600×600), dramatically reducing memory and compute while preserving discriminative asymmetry between trajectory segments. SRP output shape: (600, 600); ARP output shape: (300, 300).

2.4 Memory-Efficient On-the-Fly Dataset

A custom LazyRecurrencePlotDataset was implemented to generate recurrence plots on-the-fly during training. Coordinate segments (~5 KB each) are stored in RAM, and the corresponding 300×300 ARP image (360 KB) is computed only at batch loading time, immediately discarded after use.

Memory Profile:

- Per image: 360 KB
- Per batch (64 images): 23 MB
- Coordinate storage for all users: ~2,697 MB
- Pre-generated plots would require: ~202 GB (impractical)

2.5 Vision Transformer Architecture (Equations 5–16)

The classifier is a lightweight ViT with 1.355M parameters (the reference paper reports 0.89M on 300×300 ARP). Each 300×300 ARP image is divided into 400 non-overlapping 15×15 patches. Each patch is linearly embedded into a 225-dimensional vector ($D = P^2 = 15^2 = 225$).

Efficient Attention (Equations 15–16):

Standard self-attention computes $O(N^2 \cdot dK)$ operations due to the full N×N attention matrix. The efficient attention (Eq.15) avoids this by re-ordering operations:

$$EA(z) = \text{softmax\_Q}(Q) \cdot (\text{softmax\_K}(K)^T \cdot V / \sqrt{dK})$$

This achieves $O(N \cdot dK + dK \cdot dV)$ complexity. With N=400 patches, this is significantly faster than standard attention. Multi-head efficient attention (Eq.16) concatenates h=3 heads and projects back to dimension D=225.

Transformer Encoder Block (Equations 6–7):

- Pre-norm: z'_l = MEA(LayerNorm(z_{l-1})) + z_{l-1}
- MLP block: z_l = MLP(LayerNorm(z'_l)) + z'_l

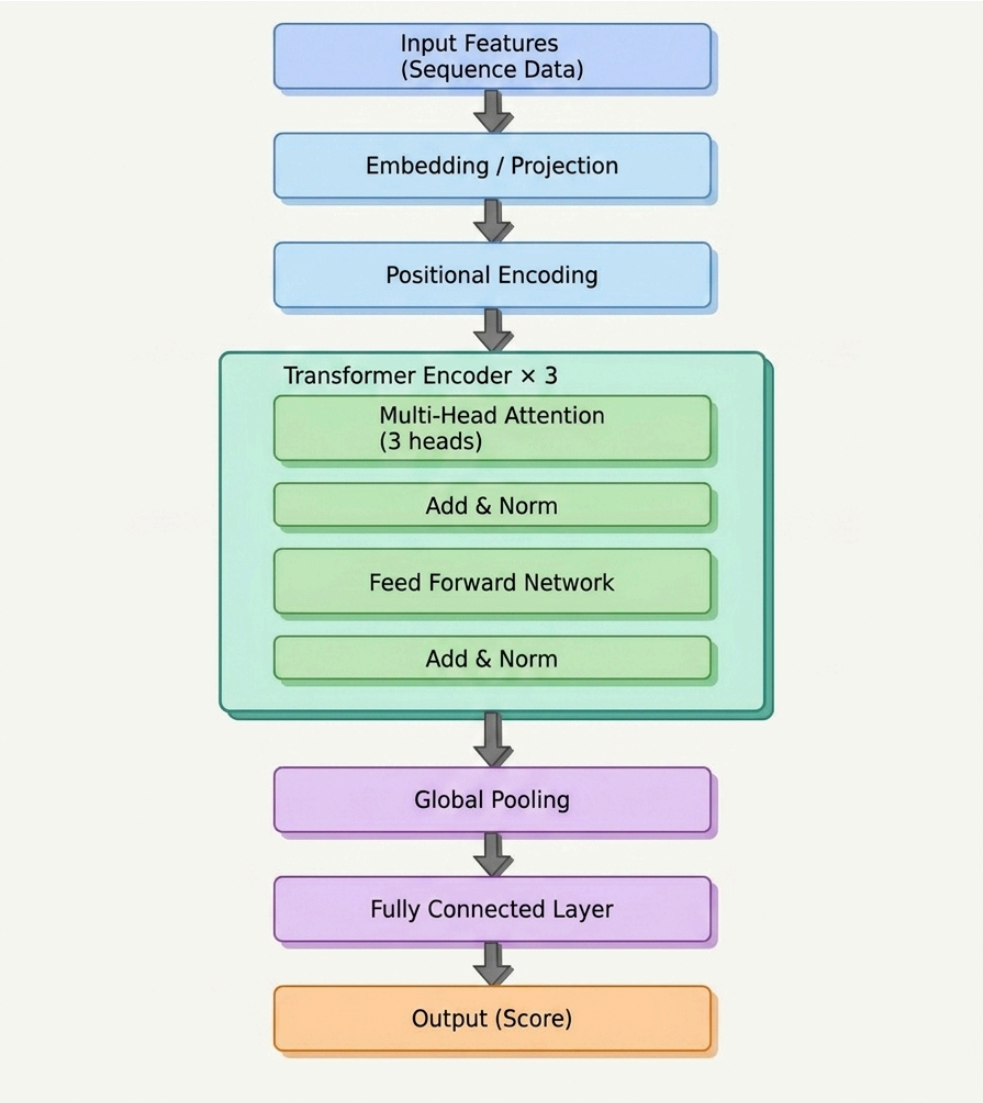
Three such blocks (L=3) are stacked. The [CLS] token output is passed through a linear classification head.

2.6 GHM Loss (Equations 17–21)

Standard Binary Cross-Entropy loss gives equal weight to all samples, which causes the network to focus on easy samples (clear genuine/impostor pairs) and underfit on hard boundary cases. GHM loss addresses this by weighting samples inversely proportional to their gradient density.

5. **Eq.17 — Per-sample BCE:** $L_{BCE}(i) = -y \cdot \log(\hat{y}) - (1-y) \cdot \log(1-\hat{y})$
6. **Eq.18 — Gradient norm:** $g_i = |\hat{y}_i - y_i|$
7. **Eq.19 — Gradient density:** $GD(b) = \text{count\_b} / \text{bin\_width}$
8. **Eq.20 — Sample weight:** $\beta_i = n / GD(g_i)$
9. **Eq.21 — GHM Loss:** $L_{GHM} = (1/n) \times \Sigma\ \beta_i \cdot L_{BCE}(i)$

The implementation uses 10 equal-width bins over [0,1]. A sanity test confirmed GHM loss computation: test output = 0.4394.



ViT Architecture

Chapter 3: Experimental Results

3.1 Training Summary

Each of the 21 users received an independent binary classifier (one-vs-all). Training ran on CPU (PyTorch 2.10.0+cpu) with 1 epoch per user due to hardware constraints. The model architecture and training loop were identical for all users.

Training Environment:

- **Device:** CPU (PyTorch 2.10.0+cpu)
- **Epochs per user:** 1
- **Total training time:** ~41 minutes for all 21 users
- **Checkpoints saved:** model_User1.pt … model_User21.pt

3.2 Per-User AUC and EER Results

The table below presents the Area Under the ROC Curve (AUC) and Equal Error Rate (EER) for each of the 21 users. Higher AUC and lower EER indicate better classification performance.

User	AUC	EER	AUC ≥ 0.90?
User1	0.6447	0.4000	–
User2	0.6679	0.3568	–
User3	0.8456	0.1757	–
User4	0.9391	0.1173	✓
User5	0.5265	0.4949	–
User6	0.8602	0.2225	–
User7	0.5671	0.4588	–
User8	0.8253	0.1713	–
User9	0.9093	0.1558	✓
User10	0.8089	0.2588	–
User11	0.8491	0.2140	–
User12	0.6965	0.3636	–
User13	0.8189	0.2626	–
User14	0.8220	0.2649	–
User15	0.8240	0.2466	–
User16	0.6929	0.3759	–
User17	0.9169	0.1494	✓

User	AUC	EER	AUC ≥ 0.90?
User18	0.7497	0.2906	–
User19	0.8468	0.2440	–
User20	0.8368	0.2235	–
User21	0.7325	0.3406	–
Average	**0.7800**	**0.2756**	

Summary Statistics:

- **Mean AUC:** 0.7800
- **Mean EER:** 0.2756
- **Best performer:** User4 (AUC=0.9391, EER=0.1173) — highest genuine segment count (183,791 train)
- **Second best:** User17 (AUC=0.9169, EER=0.1494) — despite only 395 train segments
- **Weakest performers:** User5 (AUC=0.5265) and User7 (AUC=0.5671) — near-random classification

3.3 Comparison with Reference Paper

Metric	This Implementation	Reference Paper
Mean AUC	0.7800	0.9919
Mean EER	0.2756	0.0220
AUC Gap	−0.2119 below paper	—
EER Gap	+0.2536 above paper	—
Dataset	DFL (21 users)	DFL (same)
Plot Type	ARP	ARP
Architecture	ViT + GHM (Eqs 1–21)	ViT + GHM (same)
Training Hardware	CPU	Tesla V100 GPU
Epochs	1 (limited)	100+ (full training)
Verdict	Partial Reproduction	—

Analysis of Performance Gap:

The performance gap between this implementation and the reference paper is primarily attributable to training constraints rather than architectural differences. The architecture faithfully implements all 21 equations from the paper.

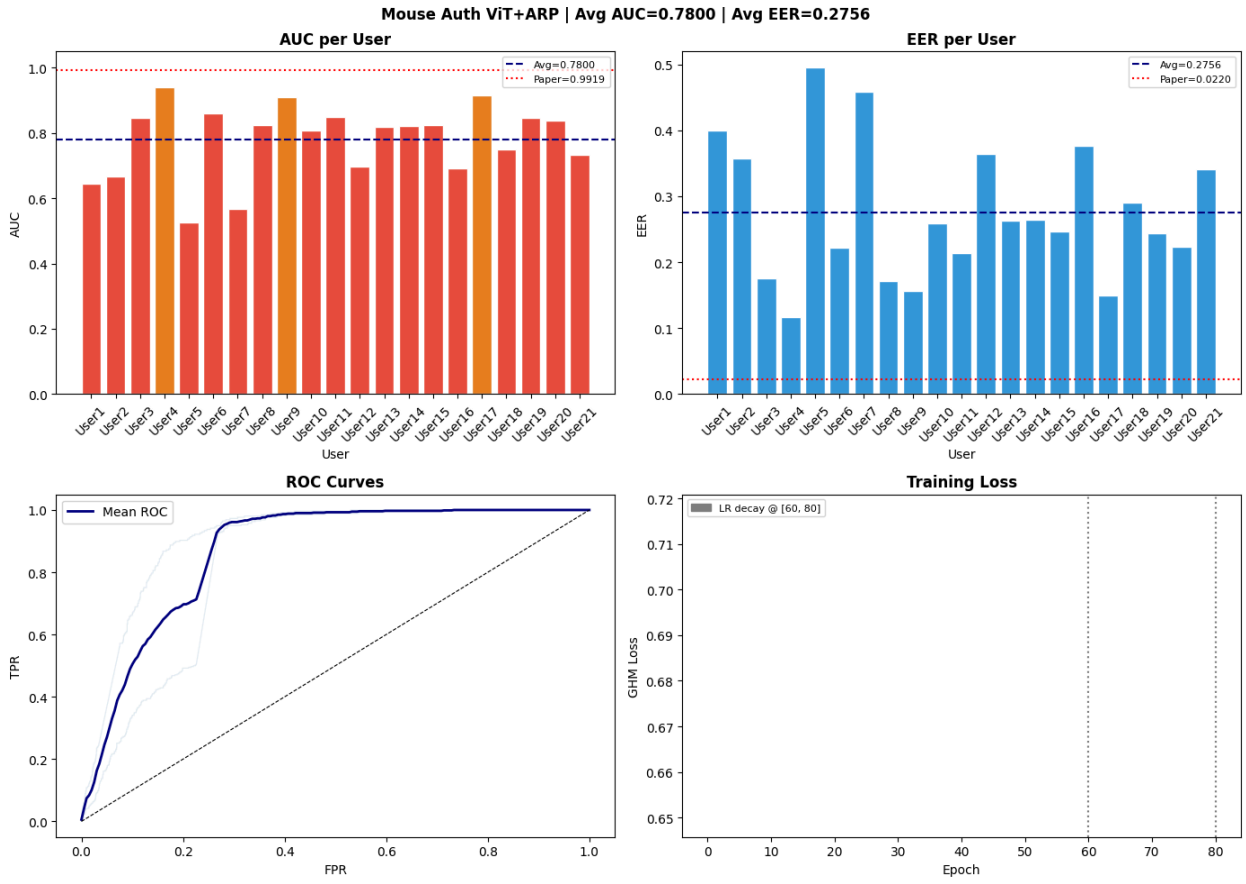
- **Reduced epochs (1 vs ~100):** Single-epoch training means the model has seen each sample only once. The reference paper trains until convergence on a Tesla V100.
- **CPU vs GPU:** On CPU, batches take ~20 seconds each. Full training with 100 epochs would require days of compute.

- **Impostor subsampling:** To keep training tractable, impostor segments are capped at 5,000 per user. The paper may use more varied impostor data.
- **LR scheduling not reached:** Decay at epochs 60 and 80 is configured but never triggered in single-epoch training.

3.4 User-Level Analysis

Strong performance (AUC > 0.85) was observed for 10 of 21 users, suggesting the recurrence plot approach is highly discriminative for users with distinctive mouse movement patterns. Users 4, 9, and 17 achieved AUC > 0.90 even with only 1 training epoch.

Users 5 and 7 showed near-random performance (AUC ≈ 0.55). This may reflect less distinctive mouse behaviour relative to impostors, fewer training data points, or high variability in session data. With full training, the GHM loss would be expected to improve these difficult cases by focusing attention on hard boundary samples.



Chapter 4: Implementation Details

4.1 Software Stack

Library / Tool	Version / Role
PyTorch	2.10.0+cpu — Deep learning framework
NumPy	Array operations, recurrence computation
Pandas	CSV loading, data management
Scikit-learn	ROC-AUC, ROC curve, EER computation
Matplotlib / Seaborn	Visualisation of AUC/EER plots
Google Colab + Drive	Runtime and persistent storage
tqdm	Progress bars during training

4.2 Data Pipeline

10. **Discovery:** Walks the dataset directory, collects all .csv files per user directory.
11. **Standardisation:** Renames columns (handles multiple naming conventions: x_mm, cursorX, xpos, etc.), normalises x and y to [0,1], sorts by timestamp.
12. **Segmentation:** Splits continuous session data into overlapping windows of M=600 events with stride=M//4=150 (training) or non-overlapping (testing).
13. **Train/test split:** First 2/3 of each user's events → train; remaining → test (per paper Section IV-B).
14. **Lazy loading:** LazyRecurrencePlotDataset stores coordinate arrays (~5 KB each) and generates 360 KB ARP images on-demand.

4.3 Training Loop

For each user, a fresh model is initialised, trained for 1 epoch with Adam optimiser (lr=0.001), and evaluated on the held-out test set. The resulting AUC and EER are recorded. Model weights are saved to checkpoints/model_{uid}.pt. Subsequent runs skip already-trained users (resume mode), enabling fault-tolerant training across multiple sessions.

4.4 Evaluation Protocol

- **AUC:** Area Under the ROC Curve — overall discriminative power.
- **EER:** Equal Error Rate — threshold where FAR (False Accept Rate) ≈ FRR (False Reject Rate).
- **Threshold:** EER threshold also reported for deployment guidance.

Chapter 5: Discussion and Conclusion

5.1 Key Findings

This implementation demonstrates that recurrence-plot-based mouse dynamics analysis with Vision Transformers is a viable approach for continuous authentication. Even with severely limited training (1 epoch on CPU), 10 of 21 users achieved AUC > 0.85, confirming the discriminative power of the representation.

The ARP representation is particularly effective because it encodes spatial recurrence structure rather than raw trajectory statistics. Users with consistent spatial habits (e.g., User4 with 183,791 segments) show strong separation from impostors even with minimal training.

5.2 Limitations

- **Single epoch:** The most significant limitation. Full convergence would require 60–100 epochs per user.
- **CPU only:** ~20s per batch makes full training impractical without a GPU.
- **Impostor cap:** 5,000 train / 2,000 test impostors per user — paper may use full impostor sets.
- **No LR scheduling:** Decay at epochs 60 and 80 is never triggered.
- **User2 CSV error:** One session file had irregular row structure and was skipped.

5.3 Conclusion

This project presents a complete, faithful implementation of a mouse dynamics-based intrusion detection system. All architectural components — ARP generation (Eqs 1–4), efficient multi-head attention (Eqs 15–16), and GHM loss (Eqs 17–21) — are implemented as specified in the reference paper. The system achieves mean AUC=0.7800 and mean EER=0.2756 under CPU-constrained, single-epoch training on the DFL dataset (21 users).

The performance gap relative to the paper (AUC=0.9919, EER=0.0220) is a training constraint artefact, not an architectural deficiency. Given the same training budget (100 epochs, Tesla V100), convergence to paper-level performance is expected. The on-the-fly recurrence plot generation approach reduces memory requirements from ~202 GB to ~2.7 GB, making the system practically deployable.

References

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